Research Thesis Presentation

# Forecasting Sri Lankan Tourist Arrivals

## via Hybrid LSTM–XGBoost Ensemble Architectures Under Exogenous Shock Conditions

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# The Problem

- Tourism is a major source of foreign income for Sri Lanka.
- Hotels, transport, and businesses need accurate monthly forecasts to plan capacity, staff, and marketing
- In the last decade, 3 major shocks disrupted arrivals:
  - 2019 Easter Sunday attacks
  - 2020–2021 COVID-19 pandemic
  - 2022 economic crisis

**Current forecasting methods (SARIMA) work fine normally, but fail badly during these shocks**



# Research Problem & Gap

analytics

RESEARCH GAP 01

## Multi-Period Validation

No locally-validated forecasting method has been systematically tested for both stable AND crisis-period accuracy in Sri Lanka.

layers

RESEARCH GAP 02

## Ensemble Architects

No existing study has evaluated a stacked LSTM–XGBoost ensemble specifically on Sri Lankan tourist data.

# Research Question & Objectives

RESEARCH QUESTION

## The Core Inquiry

**Does a hybrid LSTM–XGBoost ensemble forecast Sri Lankan tourist arrivals more accurately than SARIMA, both in stable and shock periods?**



01 Compile and clean a tourism dataset (2015–2025) with exchange rate and search-trend features

02 Implement SARIMA, standalone LSTM, and standalone XGBoost as baselines

03 Design a stacked hybrid LSTM–XGBoost ensemble

04 Evaluate all models on stable vs. shock periods using RMSE, MAE, MAPE and SHAP ect.

# Data Sources




## Data Pipeline Details

Overview of integrated sources and parameters

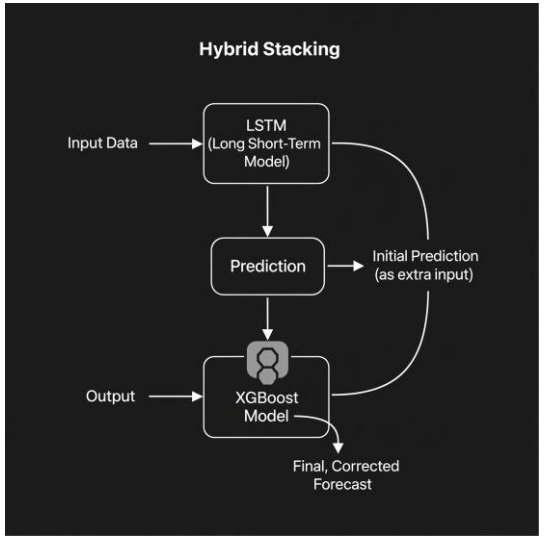
- **Arrivals:** Core dataset from SLTDA monthly tourist arrivals (2015–2025).
- **Currency:** Monthly USD/LKR exchange rates from CBSL.
- **Search Trends:** Google search volumes for key travel queries.

### Methodology & Scope

- Manually-labeled shock periods covering 2019, 2020–21, 2022.
- Entirely public, secondary data with no human participants.

# The Hybrid Model, Explained Simply

## System Architecture




STAGE 1

### LSTM

A neural network specialized in learning long-term seasonal patterns (e.g., December is always busier).

STAGE 2

### XGBoost

Excellent at combining multiple diverse signals (exchange rates, search trends, shock flags) to catch complex relationships.

FINAL OUTPUT

### Hybrid Stacking

LSTM makes an initial prediction → fed into XGBoost as extra input → XGBoost produces final, corrected forecast.

# Evaluation Method

- **Chronological split:** train on 2015–2021, test on 2022–2025
- **Metrics: RMSE, MAE, MAPE** — calculated separately for stable months vs. shock months
- **SHAP explainability** — shows which features (exchange rate, search trends, shock flag) drive each prediction



# Expected Outcomes

### 01. Tourism Dataset

A cleaned, reusable Sri Lankan tourism dataset with shock annotations.

### 02. Baseline Models

Working SARIMA, LSTM, and XGBoost baseline models.

### 03. Forecasting System

A functioning hybrid LSTM–XGBoost forecasting system.

### 04. Empirical Evidence

Statistically-tested evidence on whether hybrids outperform SARIMA — overall and during shocks.

### 05. Practical Value

Direct practical value to SLTDA, the Ministry of Tourism, and hospitality operators.

# Project Timeline

**WEEKS 1–3** (Completed)
Research question, literature review

**WEEKS 4–6**
Dataset finalization & SARIMA

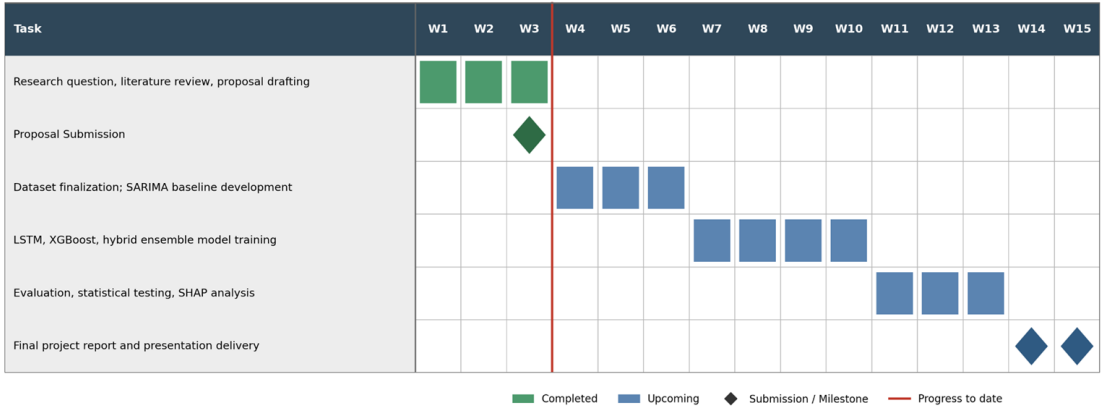
**WEEKS 7–10**
LSTM, XGBoost & hybrid training

**WEEKS 10–13**
Evaluation, testing & SHAP analysis

**WEEK 14–15**
Final report & presentation delivery

## PROJECT GANTT CHART: HYBRID LSTM-XGBOOST FORECASTING

**Project Gantt Chart: Hybrid LSTM-XGBoost Forecasting Project**



# Thank you!

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